ference between baseline semiconductors and the semiconductors that have AI functionality.

At present, about 10% of the chips being produced have some sort of AI in them. Notable examples are cellphone application processors that have AI engines to do things like facial recognition. About half of the market will be represented by chips with AI by 2030.

Explosion in global funding

The first thing to look at is venture capital (VC) investments (see Fig. 6). Before the dot-com crash, the VC investments were about 2.5 billion dollars, which dropped by about 40% after the crash to about 1.7 billion dollars for a few years. In 2009, the great recession dropped that down further to under 1 billion dollars a year, which were being spent on semiconductor content. It continued to fall until 2015, when it eventually dropped to half a billion dollars. Thereafter, the investments picked up sharply and in 2021 we had almost 10 billion dollars investment flowing into startup companies in the semiconductor space.

Previously, the majority of the spending was being done in the United States (US) alone but now this trend has spread globally. China, with its well-known effort to become an important provider in the semiconductor space, has led with almost 6.5 billion dollars of spending. But even the rest of the world, outside of the US and China, has made about 1.6 billion dollars of investment into those semiconductor companies. This trend is expected to increase further as more and more nations are realising the strategic importance of semiconductors for enabling their overall economic growth.

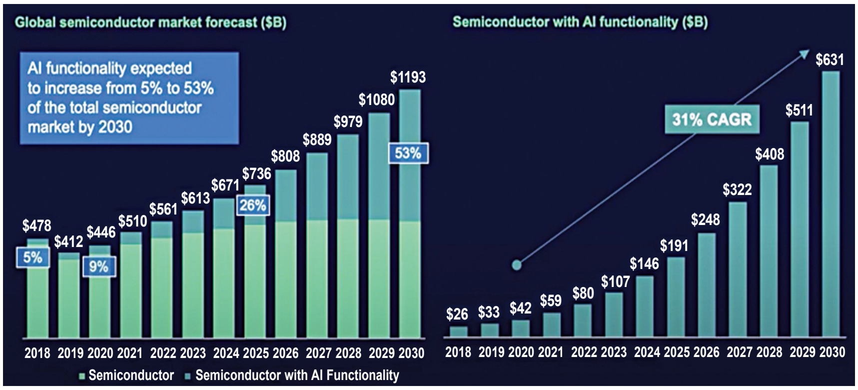

Fig. 5: Global semiconductor market forecast

Advanced logic and foundry capital investments

The chart in Fig. 7 shows the distribution of advanced logic and foundry capital expenditure. It can be seen that was a relatively flat spending until about 2018. It started increasing from the year 2019 and reached close to 80 billion dollars in 2021. Taiwan Semiconductor Manufacturing Company (TSMC) has already stated that they will spend approximately 40 billion dollars on enabling new foundry capacity next year. So, we can expect to see a significant increase in this number in the future.

Announcement by the Indian government

The government of India announced approximately 10 billion dollars of incentives for the electronics and semiconductor industry on 21st December last year, which was particularly encouraging for

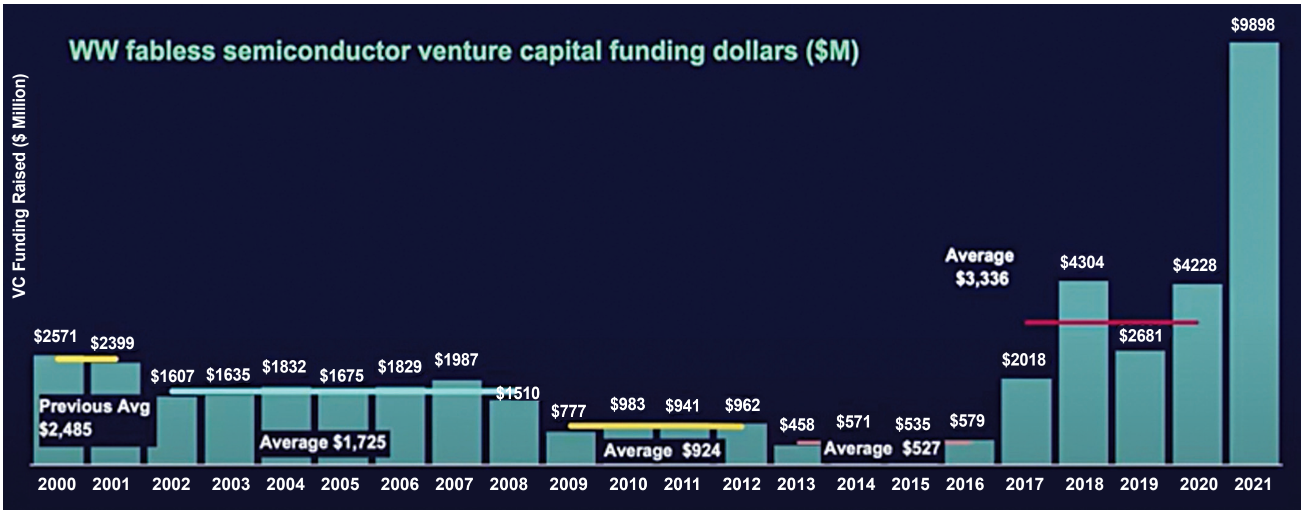

Fig. 6: Worldwide fabless semiconductor venture capital funding in millions of dollars